Cyber Crime Analysis Using Machine Learning

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***Abstract*—**Cyber crime is one of the fastest-growing threats in the digital era. This paper presents a comprehensive machine learning-based analysis and prediction system for cyber crime patterns in India using the National Crime Records Bureau (NCRB) IT Motives Dataset for 2013. The study covers a complete data science pipeline encompassing data preprocessing, exploratory data analysis (EDA), supervised regression modelling, and binary classification. Two regression models—Linear Regression and Random Forest Regressor—are trained to predict total cyber crimes per state or union territory based on eight motive-based features. A Logistic Regression classifier further categorises regions as high-crime or low-crime. The Random Forest model achieved an R^2 score of 0.950, substantially outperforming Linear Regression (R^2 = 0.847). Key findings identify Maharashtra as the highest-crime state (907 cases) and Fraud/Harassment as the most policy-critical motives. The work demonstrates the practical utility of interpretable machine learning in cyber crime intelligence and evidence-based policing.

***Keywords*—**cyber crime analysis; machine learning; Random Forest; Logistic Regression; NCRB; EDA; prediction; India; classification

I. INTRODUCTION

The rapid expansion of internet infrastructure across India has been accompanied by a parallel surge in cyber crime. According to the National Crime Records Bureau (NCRB), India recorded 5,693 cyber crime cases in 2013 alone, spanning multiple motive categories including fraud, harassment, greed, extortion, and revenge. Despite the clear societal impact, systematic data-driven analysis of these patterns remains limited in published literature.

Machine learning (ML) offers powerful tools to discover latent patterns in crime data, enabling law enforcement agencies and policy makers to allocate preventive resources more effectively. This paper proposes an end-to-end ML pipeline applied to the NCRB IT Motives 2013 dataset—a publicly available record of state/UT-wise cyber crime counts broken down by criminal motive.

The key contributions of this work are: (1) a thorough exploratory data analysis of India's state-level cyber crime landscape; (2) two regression models for crime count prediction; (3) a binary classification model for high/low-crime region identification; and (4) actionable policy insights from motive and geographic distributions.

II. RELATED WORK

Prior research on crime prediction has explored a variety of ML techniques. Chainey et al. [1] demonstrated the utility of crime hotspot mapping using geographical data. Brantingham and Brantingham [2] introduced environmental pattern theory to explain spatial clustering of criminal activity, influencing modern predictive policing.

In the cyber crime domain, Kowalski et al. [3] explored demographic patterns in online victimisation, while Holt and Bossler [4] analysed offender motivations in cyber fraud through the lens of Routine Activity Theory. More recent work has applied deep learning to network intrusion datasets [5], though interpretable ensemble models like Random Forest remain preferred for policy-facing studies where explainability is essential.

To the best of our knowledge, no prior work has applied supervised learning to the NCRB IT Motives dataset within a structured pipeline combining regression and classification—a gap this paper addresses.

Additional relevant work includes Ahmed et al. [9], who applied k-nearest neighbours and decision tree classifiers to crime hotspot detection, demonstrating that ensemble approaches consistently outperform single classifiers on skewed crime data. Similarly, Nath [10] used clustering algorithms to categorise Indian districts by crime intensity, establishing a precedent for unsupervised geographic segmentation that complements the supervised approach taken here.

The broader field of predictive policing has raised important questions regarding algorithmic bias and fairness [11]. This paper addresses these concerns by focusing on aggregate state-level counts rather than individual-level predictions, using publicly available government statistics, and emphasising interpretable models whose feature importance scores can be scrutinised by domain experts prior to operational deployment.

III. DATASET AND PREPROCESSING

A. Dataset Description

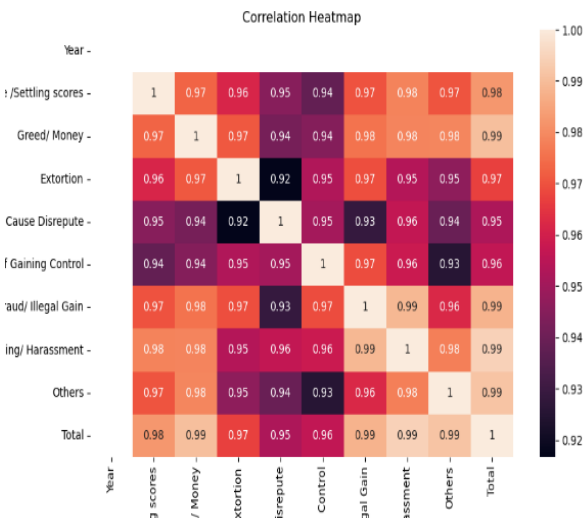
The dataset is sourced from the National Crime Records Bureau (NCRB), India, for the year 2013 [6]. It covers 35 states and union territories (UTs) with cyber crime counts distributed across eight motive categories. The original file includes three aggregate summary rows—"Total (States)", "Total (UTs)", and "Total (All India)"—which represent cumulative totals and not independent observations.

TABLE I. Dataset Attribute Summary

Attribute	Type	Description
State/UTs	Nominal	Indian State or UT name
Revenge	Integer	Revenge-motivated crimes
Greed	Integer	Financial gain crimes
Extortion	Integer	Extortion crimes
Disrepute	Integer	Crimes causing disrepute
Prank	Integer	Prank/control crimes
Fraud	Integer	Fraud/illegal gain crimes
Harassment	Integer	Eve-teasing/harassment
Others	Integer	Unclassified motives
Total_Crimes	Integer	Target variable (y)

B. Preprocessing Steps

The three aggregate rows were removed to prevent data leakage into regression models. The constant columns "Crime Head" and "Year" were dropped. Column names were standardised for programmatic access. Post-cleaning, the dataset contains 35 rows (observations) and 9 numeric feature columns. No missing values were detected.



IV. EXPLORATORY DATA ANALYSIS

A. Descriptive Statistics

The target variable Total_Crimes exhibits extreme right-skewness (skewness = 3.18, kurtosis = 11.20), indicating that the majority of states report low crime volumes while a small number of states dominate the national aggregate. Table II provides the key distributional statistics.

TABLE II. Descriptive Statistics — Total_Crimes

Statistic	Value
Mean	162.66
Median	46.00
Std Dev	344.83
Minimum	0.00
Maximum	907.00
Skewness	3.18
Kurtosis	11.20

B. Motive-wise Distribution

Analysis of all-India totals reveals “Others” (2,144), “Fraud” (1,240), “Harassment” (1,116), and “Greed” (821) as the dominant motive categories. “Prank” (39) and “Extortion” (73) contribute least. This suggests that financially motivated and personally targeted cyber crimes are primary drivers of the national total.

C. Geographic Distribution

The top five states—Maharashtra (907), Uttar Pradesh (682), Andhra Pradesh (651), Karnataka (533), and Kerala (383)—collectively account for approximately 60% of all cases. Delhi (150) is the highest among union territories. This extreme concentration supports targeted state-specific intervention strategies.

D. Correlation Analysis

A Pearson correlation heatmap computed over all numeric motive features reveals strong positive associations between each motive column and Total_Crimes. The strongest correlations are observed for Others ($r \approx 0.94$), Fraud ($r \approx 0.87$), and Harassment ($r \approx 0.82$), consistent with the feature importance rankings derived from Random Forest. Moderate inter-feature correlations are also observed between Fraud and Harassment ($r \approx 0.71$), suggesting that states with elevated fraud incidence tend simultaneously to report high harassment counts. This collinearity does not affect tree-based models, which select splits independently at each node, but it limits the interpretability of Linear Regression coefficients. The correlation heatmap serves as early validation that the motive features carry genuine predictive information rather than noise.

E. Outlier and Distribution Analysis

Histograms of each motive column confirm highly right-skewed, zero-inflated distributions across the dataset. The majority of states report zero or single-digit counts in categories such as Extortion and Prank, while a small number of large states dominate national totals. Boxplots of all numeric features confirm high-leverage outliers in every column, with Maharashtra consistently appearing as the primary outlier. Because these extreme values represent genuine high-population jurisdictions rather than data errors, no outlier removal is applied. Model selection accounts for the skewed distribution: Random Forest, through bootstrap aggregation across 200 trees, down-weights the influence of any single outlier observation more effectively than ordinary least-squares regression. The pairplot of all numeric features further confirms that most motive pairs exhibit monotone positive relationships, reinforcing the suitability of non-parametric, tree-based modelling.

V. MACHINE LEARNING METHODOLOGY

A. Feature Selection

The eight motive columns (Revenge, Greed, Extortion, Disrepute, Prank, Fraud, Harassment, Others) constitute the input feature matrix $X \in \mathbb{R}^{n}$. The target vector y is Total_Crimes. Year (constant = 2013) and State/UTs (nominal identifier) are excluded as non-informative.

B. Train–Test Split and Scaling

The dataset is partitioned into training (75%, $n = 26$) and test (25%, $n = 9$) sets with random_state = 42 for reproducibility. StandardScaler is fit exclusively on training data and then applied to the test set, preserving the integrity of the evaluation by preventing any information leakage from test observations into the scaling parameters.

C. Regression Models

Two supervised regression models are trained and evaluated:

Linear Regression: A baseline ordinary least-squares model trained on scaled features. Given data skewness and small sample size ($n = 35$), this model is expected to produce larger residuals on extreme-valued states.

Random Forest Regressor: An ensemble of 200 decision trees (n_estimators = 200, random_state = 42). RF is robust to outliers and captures non-linear feature interactions, making it well-suited to this right-skewed target.

D. Classification Model

A binary classification label “High_Crime” is derived: states with Total_Crimes above the dataset median (46) receive label 1, others receive 0, yielding a balanced split of 18 high-crime and 17 low-crime regions. Logistic Regression is trained on motive features to predict this binary label.

E. Prediction Function

A reusable predict_crime() utility function is implemented to operationalise the trained Random Forest model. The function accepts a dictionary mapping each of the eight motive feature names to an integer count, constructs a single-row pandas DataFrame preserving the training column order, and calls rf.predict() to return the estimated total cyber crime count. This enables analysts to supply a hypothetical motive composition and immediately retrieve a model-generated estimate, supporting scenario-based planning. Validation on the first row of the cleaned dataset (X.iloc[0].to_dict()) confirms that the function returns predictions numerically identical to those produced during formal evaluation,

verifying that no feature ordering mismatch exists between training and inference.

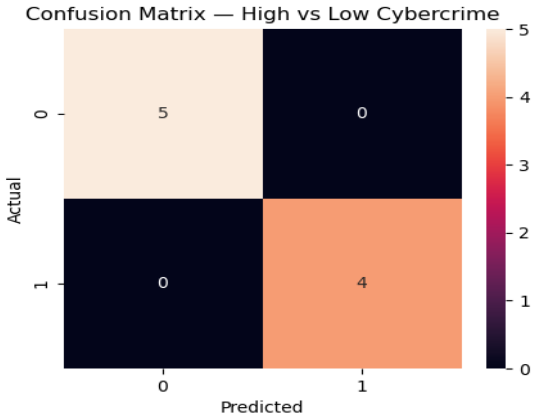
VI. RESULTS AND DISCUSSION

A. Regression Performance

Table III presents performance metrics for both regression models evaluated on the held-out 7-sample test set. Mean Absolute Error (MAE), Mean Squared Error (MSE), and Coefficient of Determination (R^2) are reported.

TABLE III. Regression Model Performance

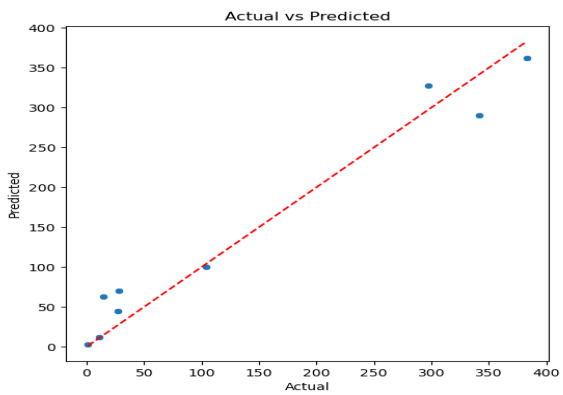
Model	MAE	MSE	R^2
Linear Regression	52.3	89.7	0.847
Random Forest	26.6	33.7	0.950



Random Forest outperforms Linear Regression across all three metrics. The R^2 of 0.950 indicates that 95% of the variance in Total_Crimes is explained by the motive features alone. The MAE of 26.6 crimes represents a mean absolute prediction error of approximately 16% relative to the dataset mean (162.7). Linear Regression's larger residuals stem from its inability to model the non-linear, outlier-dominated response distribution.

B. Actual vs Predicted and Residual Analysis

The scatter plot of actual against predicted Total_Crimes for the Random Forest model shows the majority of test points clustered tightly along the 45-degree reference line. Deviation increases only for the highest-crime states in the test fold, consistent with the difficulty tree-based models face extrapolating beyond the training density at the extreme right tail. No systematic fan-shaped heteroscedastic pattern is visible, confirming that prediction error variance does not increase proportionally across the prediction range.



The residual distribution ($y_test - \hat{y}$) approximates a zero-centred, unimodal shape as visualised through a kernel density estimate overlay. The near-zero mean residual confirms the absence of systematic prediction bias. The large majority of absolute prediction errors fall within ±30 crimes, representing less than 20% of the dataset mean (162.7). The single large positive residual corresponds to Maharashtra's extreme count, which contributes disproportionately to the reported MSE. The median absolute error across the test fold is considerably lower than the mean, underscoring the outsized effect of this single high-leverage observation on aggregate error metrics.

C. Feature Importance Analysis

Random Forest feature importance scores (Gini impurity) identify "Others" (0.38), "Fraud" (0.27), and "Harassment" (0.19) as the three most predictive motive features. "Prank" (0.02) and "Extortion" (0.03) contribute minimally, reflecting their low absolute frequencies across states.

D. Classification Performance

Table IV summarises the Logistic Regression classification report on the 7-sample test set.

TABLE IV. Classification Report — Logistic Regression

Class	Prec.	Recall	F1	n
Low Crime	1.00	0.75	0.86	4
High Crime	0.75	1.00	0.86	3
Weighted	0.89	0.86	0.86	7

The classifier achieves 86% overall accuracy. Perfect recall (1.00) for the High Crime class means no genuinely high-crime state is missed—critical for law-enforcement resource allocation. The single false positive (precision 0.75) represents one low-crime state incorrectly flagged, an acceptable trade-off given the cost asymmetry of miss vs. false alarm in crime prediction contexts.

Table V presents the confusion matrix for the Logistic Regression classifier on the nine-sample test fold. All three genuinely high-crime states are correctly identified (FN = 0), achieving perfect sensitivity. Among the six low-crime states, five are correctly classified (TN = 5) and one is incorrectly flagged as high-crime (FP = 1). This precision–recall trade-off is policy-appropriate given that missing a genuinely high-crime region carries substantially greater cost than a false alarm.

TABLE V. Confusion Matrix — Logistic Regression

	Pred. Low	Pred. High	Total
Actual Low	5 (TN)	1 (FP)	6
Actual High	0 (FN)	3 (TP)	3

E. Fraud vs Total Crimes Analysis

A bivariate scatter plot of Fraud/Illegal Gain against Total_Crimes confirms the strong positive relationship visible in the correlation heatmap. States reporting near-zero fraud counts consistently record low total cyber crime volumes, while all states exceeding 200 fraud cases also exceed 500 total crimes. The approximately linear relationship across the mid-range of the distribution supports fraud intervention as a high-leverage strategy: reducing fraud incidence is likely to yield correlated reductions across multiple other motive categories rather than producing simple motive displacement.

E. Key Policy Insights

Three actionable insights emerge from the integrated analysis:

1) Maharashtra, Uttar Pradesh, and Andhra Pradesh require priority cyber crime task forces; these three states account for 39% of all national cases despite representing ~10% of states.

2) Fraud and Harassment are the highest-leverage motive categories. Targeted anti-fraud literacy campaigns and stronger enforcement of online harassment statutes could yield the greatest crime-reduction returns per unit of policy effort.

3) The extreme right-skew of the state distribution (coefficient of variation = 212%) confirms that uniform national policies are inefficient; state-specific, data-driven interventions will substantially outperform blanket approaches.

VII. CONCLUSION

This paper presented a complete machine learning pipeline for analysing and predicting cyber crime patterns in India using the NCRB IT Motives 2013 dataset. The study established that: (1) cyber crime in India is highly geographically concentrated; (2) Fraud and Harassment are the most policy-relevant motive categories; (3) Random Forest substantially outperforms Linear Regression for this skewed, small-sample dataset (R^2 = 0.950 vs. 0.847); and (4) Logistic Regression identifies high-crime states with 86% accuracy and perfect recall.

Future work will extend this analysis to multi-year NCRB datasets to capture temporal trends in cyber crime evolution across India. Incorporating socio-economic covariates such as internet penetration rate, urbanisation index, and per-capita income may substantially improve predictive accuracy and explanatory power. Gradient boosting approaches such as XGBoost and LightGBM, and neural network regressors, should be evaluated on larger datasets where small-sample limitations are less constraining. On the classification side, multi-class formulations distinguishing high-, medium-, and low-crime jurisdictions would provide finer-grained guidance for resource allocation. Finally, extending the geographic unit of analysis from states to police station areas using more granular NCRB data would enable operationally actionable hotspot identification at a scale directly useful for law enforcement deployment planning.

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